Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Main Topic: [Banana] – [Planning and Soil Testing]

1.1 Site Selection, Soil Suitability, Testing, and Pre-Planting Field Preparation

1.1.1 1. Site Selection and Climatic Suitability

Banana (*Musa spp.*) is a tropical crop that prefers tropical humid lowlands and is cultivated from sea level up to an elevation of 1,500 m to 2,000 m above mean sea level *[Source: Banana_disease_detection.pdf, Section 2, pp. 1–2]*. In hilly regions (such as the Palney and Shevaroy hills), cultivation requires clearing jungle vegetation and constructing contour stone walls along slopes prior to planting *[Source: Banana Expert System, Field Preparation]*4.

Temperature and Climate Limits The crop grows optimally within a temperature range of 15°C to 35°C, with a relative humidity of 75% to 85% *[Source: Banana_disease_detection.pdf, Section 1, p. 1]*1. The optimum temperature for leaf emergence is 25°C–30°C, with absolute physiological limits ranging from 10°C to 40°C *[Source: Banana_disease_detection.pdf, Section 2, pp. 1–2]*3. Chilling injury occurs at temperatures below 12°C *[Source: Banana_disease_detection.pdf, Section 2, p. 1]*2. Planting should be avoided during cold weather (below 15°C), extreme summer heat (above 40°C), or periods of heavy rainfall when soils remain excessively wet *[Source: Banana Expert System, TC Do's and Don'ts #6]*5.

Rainfall, Water Availability, and Wind Protection A four-month monsoon period (June to September) providing an average rainfall of 650–750 mm is critical for vigorous vegetative growth *[Source: Banana_disease_detection.pdf, Section 2, p. 1]*2. For optimum growth, monthly rainfall should be evenly distributed at 20–22 mm; areas with distinct long dry seasons require assured irrigation facilities *[Source: Banana_disease_detection.pdf, Section 2, pp. 1–2]*3. Total crop water requirement over its life cycle is 900–1,200 mm *[Source: Banana_disease_detection.pdf, Section 7, p. 7]*6. Wind speeds exceeding 30 km/hr cause lodging in shallow-rooted banana plants, while velocities above 80 km/hr cause severe structural destruction *[Source: Banana_disease_detection.pdf, Section 2, p. 1; Banana Expert System, Contingency Problems]*27. Windbreaks or wind shields such as Shevri (*Sesbania egyptifolia*), Dhaincha (*Sesbania aculeata*), or Sunnhemp (*Crotalaria juncea*) should be established along field borders in an East–West orientation to regulate humidity and buffer high-velocity winds *[Source: Banana Expert System, Intercultural Operations / TC Do's #5]*89.

1.1.2 2. Soil Suitability Requirements and Characteristics

Banana can be grown across diverse soil types provided there is good drainage, adequate fertility, and sufficient moisture-retention capacity *[Source: Banana Expert System, Soil; Banana_disease_detection.pdf, Section 2, p. 1]*210.

Preferred Soil Textures and Characteristics The most preferred soils are deep, rich loams, silty clay loams, and organically rich sandy loams with good internal drainage and aeration *[Source: Banana Expert System, Soil / TC Do's #3; Banana_disease_detection.pdf, Section 2, p. 1]*. Soils must be rich in organic material, with high nitrogen content, adequate phosphorus levels, and abundant potash *[Source: Banana Expert System, Soil]*10.

Unsuitable Soil Types and Constraints Ill-drained, poorly aerated, and nutritionally deficient soils are unsuitable *[Source: Banana Expert System, Soil]*10. Cultivation must be avoided in extreme clayey soils, very sandy soils, saline soils, calcareous soils, low-lying waterlogged areas, and heavy black cotton soils with poor drainage *[Source: Banana Expert System, Soil]*10.

Soil Reaction (pH), Electrical Conductivity (EC), and Organic Carbon

- **Soil pH Requirements and Disagreements:** *Banana Expert System* specifies the ideal soil pH range as 6.0 to 7.5, noting that soils exceeding pH 7.5 are unsuitable without amendment *[Source: Banana Expert System, Soil / TC Do's #3]*. Conversely, *Banana_disease_detection.pdf* specifies the most preferred pH range as 6.5 to 7.5 *[Source: Banana_disease_detection.pdf, Section 2, p. 1]*2. High soil pH (> 7.5 or > 8.5) induces micro-nutrient tie-up (such as iron chlorosis) *[Source: Banana Expert System, Calendar 6th Month; Banana_disease_detection.pdf, Section 6.1]*1213.
- **Electrical Conductivity (EC) and Organic Carbon Status:** Exact numerical ranges for Electrical Conductivity (EC in dS/m) or organic carbon percentages are not explicitly quantified in the provided sources. However, sources mandate that soils must not be saline or alkaline and must not be devoid of organic matter *[Source: Banana Expert System, Soil / TC Do's #3–4; Banana.pdf, p. 4]*.

1.1.3 3. Pre-Planting Soil Testing, Sampling, and Analysis Procedures

Soil testing prior to plantation establishment is mandatory to assess major and minor nutrient statuses, identify deficiencies or toxicities, measure soil reaction/salinity, and detect pathogenic fungi, bacteria, and plant-parasitic nematodes *[Source: Banana Expert System, Soil / TC Do's #2; Banana.pdf, p. 4; Banana_disease_detection.pdf, Section 11, p. 38]*.

Procedure for Soil Sampling and Nematode Extraction For diagnosing plant-parasitic nematodes (such as *Radopholus similis*, *Pratylenchus coffeae*, and *Meloidogyne* spp.) prior to planting, representative soil samples must be collected and processed using standardized extraction protocols *[Source: Banana.pdf, Section II.F, p. 11]*17:

1. **Sample Collection:** Collect a 100 to 300 cm^3 (200–300 g) representative soil sample from the root zone area *[Source: Banana.pdf, Section II.F, p. 11]*17.
2. **Sample Preparation:** Thoroughly mix the soil sample and pass it through a coarse sieve to remove rocks, debris, and roots *[Source: Banana.pdf, Section II.F, p. 11]*17.

3. **Subsampling and Agitation:** Measure a 600 cc subsample of soil, pack it lightly and uniformly into a beaker, and transfer it into a vessel half-filled with water. Stir the soil-water mixture with a paddle and allow it to settle until swirling ceases *[Source: Banana.pdf, Section II.F, p. 11]*17.

4. **Sequential Sieving:**

   - Pour liquid through a 20-mesh sieve into a second bucket; discard the heavy sediment from the first bucket and material caught on the sieve *[Source: Banana.pdf, Section II.F, p. 11]*1718.
   - Stir the second bucket, allow it to settle, and pour through a 200-mesh sieve. Backwash the material caught on the 200-mesh sieve (containing large nematodes) into a 250-ml beaker *[Source: Banana.pdf, Section II.F, p. 11]*1718.
   - Stir liquid in the first bucket, allow settling, and pour through a 325-mesh sieve. Backwash the material caught on the 325-mesh sieve (containing small to mid-sized nematodes and silt) into the 250-ml beaker *[Source: Banana.pdf, Section II.F, p. 11]* [31, 158–159].

5. **Recovery:** Over 90% of live nematodes are recovered in the first 5–8 mm of water drawn through rubber tubing into a shallow dish for microscopic examination *[Source: Banana.pdf, Section II.F, p. 11]*19.

Agro-Ecosystem Analysis (AESA) Root and Soil Scouting Under AESA guidelines, field scouting requires checking unhealthy plants and examining root systems for lesions, galls, or cortical decay. Roots should be washed in water and split longitudinally to inspect for internal discolouration caused by fungal or bacterial pathogens *[Source: Banana.pdf, p. 9; Banana_disease_detection.pdf, Section 10.3, p. 10]*2021.

1.1.4 4. Soil Test Results, Deficiency Symptoms, and Corrective Measures

Nutrient applications must be strictly based on soil test recommendations; over-application of nitrogen causes excessive succulence and pest/disease susceptibility, whereas under-application retards growth *[Source: Banana.pdf, p. 4, 38; Banana_disease_detection.pdf, Section 11, p. 38]*. Generally, a banana crop requires 7–8 kg N, 0.7–1.5 kg P, and 17–20 kg K per metric ton of fruit yield *[Source: Banana.pdf, p. 14; Banana_disease_detection.pdf, Section 6, p. 5]*23.

Multi-Micronutrient Formulation Combined foliar application of $ZnSO_4$ (0.5%), $FeSO_4$ (0.2%), $CuSO_4$ (0.2%), and H_3BO_3 (0.1%) applied at 3, 5, and 7 months post-planting enhances yield *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*25. Alternatively, spray ICAR-NRCB banana micronutrient mixture @ 10 g/L during the 4th, 5th, 6th, and 7th months with a surfactant *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*27.

1.1.5 5. Pre-Planting Field Preparation, Tillage, and System-Specific Methods

Primary Tillage and Summer Solarization The land must be ploughed 2 to 4 times (or 4 to 6 times in garden lands) and leveled *[Source: Banana Expert System,*

Table 1: Soil-Test Identified Nutrient Deficiencies, Diagnostic Symptoms, and Source-Recommended Corrective Actions

Nutrient	Diagnostic Deficiency Symptoms	Recommended Corrective Measures
Nitrogen (N)	Retarded crop growth; excessive dosage causes crop succulence and susceptibility to pests/diseases *[Source: Banana.pdf, p. 4]*1422.	Apply 25% N in organic form (FYM/neem cake) and 75% in inorganic form in 3 equal split doses (60, 90, 120 days post-planting) *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*2425.
Phosphorus (P)	Stunted growth, poor root development, saw-tooth marginal chlorosis, leaf curling, petiole breakage, bluish-green young leaves *[Source: Banana_disease_detection.pdf, Section 6.1, p. 6]*26.	Soil application of DAP @ 50 g/plant followed by irrigation, or 250 g Superphosphate/plant as basal dose *[Source: Banana_disease_detection.pdf, Section 6.1, p. 6]*. In acidic soils, TSP or DAP is recommended *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*25.
Potassium (K)	Orange-yellow chlorosis of old leaves, marginal scorching, reduced leaf area, midrib curving, leaf choking delaying flowering *[Source: Banana_disease_detection.pdf, Section 6.1, p. 6]*26.	Soil application of 100–150 g MOP/plant or foliar spray of 1.0–1.5% Potassium Sulphate (K_2SO_4) solution *[Source: Banana_disease_detection.pdf, Section 6.1, p. 6]*26. Total K requirement: 300–400 g K_2O/plant in 3–5 splits *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*25.
Calcium (Ca)	Deformation or absence of leaf lamina (spike leaf), marginal leaf necrosis, vein thickening *[Source: Banana_disease_detection.pdf, Section 6.1, p. 7]*29.	Soil application of Lime ($CaCO_3$) or Dolomite ($MgCO_3$) @ 50 g/plant followed by irrigation *[Source: Banana_disease_detection.pdf, Section 6.1, p. 7]*29.
Magnesium (Mg)	Yellow discolouration in mid-blade and midrib (margins remain green), purple petiole mottling, marginal necrosis, leaf sheath separation *[Source: Banana Expert System, Calendar 5th Month; Banana_disease_detection.pdf, Section 6.1, p. 6]*2630.	Soil application of Magnesium Sulphate ($MgSO_4$) @ 25 g/plant followed by irrigation *[Source: Banana Expert System, Calendar 5th Month; Banana_disease_detection.pdf, Section 6.1, p. 6]*2630.
Sulphur (S)	Yellow/white appearance of young leaves, necrotic margin patches, vein thickening, stunted growth, small/choked bunches *[Source: Banana_disease_detection.pdf, Section 6.1, p. 7]*1326.	Soil application of Bentonite Sulphur @ 30 g/plant at 3rd or 5th month post-planting followed by irrigation *[Source: Banana_disease_detection.pdf, Section 6.1, p. 7]*613.
Iron (Fe)	Younger leaves turn yellow or white (iron chlorosis), severe in pH > 8.5 and calcareous4 soils *[Source: Banana Expert System, Calendar 6th Month; Banana_disease_detection.pdf, Section 6.1, p. 7]*	Foliar spray of 0.5% Ferrous Sulphate ($FeSO_4$) + 1.0% Urea with wetting agent *[Source: Banana Expert System, Calendar 6th Month; Banana_disease_detection.pdf, Section 6.1, p. 7]*

*Field Preparation; Banana_disease_detection.pdf, Section 4, p. 3]*432. Deep ploughing should be executed on bright sunny days during May and June, keeping the field exposed to sunlight for 2 to 3 weeks without planting or irrigation *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38].* This desiccation kills perennial weed bulbs/rhizomes, exposes white grubs, and suppresses soil-borne pathogens and nematodes *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38].*

Clod Breaking and Laser Land Leveling A rotavator or disc harrow is used to break soil clods to a fine tilth *[Source: Banana Expert System, Field Preparation; Banana_-disease_detection.pdf, Section 4, p. 3]*432. Field leveling should be completed using a blade harrow or Laser Leveler *[Source: Banana Expert System, Field Preparation]*4. Laser land leveling uses a guided laser beam to level topography within precise slope degrees, conserving irrigation water, improving nutrient-use efficiency, evening out yield variations, and enabling precision farming *[Source: Banana Expert System, Field Preparation]*4.

Organic Matter Incorporation and Green Manuring During land preparation, a basal dose of Farm Yard Manure (FYM) @ 50 tonnes/ha is applied before the final harrowing and incorporated into the soil *[Source: Banana Expert System, Field Preparation; Banana_disease_detection.pdf, Section 4, p. 3]*432. Prior to planting, green manure crops such as Sunnhemp (*Crotalaria juncea*), Dhaincha (*Sesbania aculeata*), or Cowpea should be sown and ploughed back into the soil at the flowering stage ($\sim$ 40 days after sowing) for soil conditioning *[Source: Banana Expert System, Field Preparation / Calendar 2nd Month; Banana_disease_detection.pdf, Section 4, p. 3].*

System-Specific Preparation Procedures

- **Garden Lands:** Requires 2–4 ploughings followed by pit or furrow preparation *[Source: Banana Expert System, Field Preparation]*4.
- **Wetlands (e.g., Cauvery Delta, Tamil Nadu):** No dry preparatory cultivation is conducted. The field is puddled with abundant water using a gauge wheel (like paddy land) and drained for one day *[Source: Banana Expert System, Field Preparation / Trench Planting; Banana_disease_detection.pdf, Section 5.4, p. 4].* Suckers are pressed directly into wet mud. After one week, 15 cm deep trenches are opened both ways between 4–6 plant blocks. Trenches are deepened monthly by 20–25 cm up to 60 cm depth to serve as drainage channels during monsoon and recycled for manure *[Source: Banana Expert System, Trench Planting]*38.
- **Padugai Lands:** Prepared via one deep spade digging *[Source: Banana Expert System, Field Preparation]*4.
- **Hill Banana Lands:** Prepared by clearing jungle growth and constructing contour stone walls along slopes to prevent erosion before digging pits *[Source: Banana Expert System, Field Preparation]*4.

Pitting Procedures and Source Disagreements on Pit Dimensions Pits are dug sufficiently ahead of planting at intersecting row points *[Source: Banana Expert System, Field Preparation]*4.

- **Disagreement on Pit Dimensions:** *Banana Expert System (Field Preparation)* specifies standard pit dimensions as 45 cm × 45 cm × 45 cm *[Source: Banana Expert System, Field Prep]*4. *Banana Expert System (Pit Method)* and intersecting furrow guidelines state pit sizes of 60 cm × 60 cm × 60 cm (0.6 m × 0.6 m × 0.6 m) *[Source: Banana Expert System, Pit Method]*439. *Banana_disease_detection.pdf* specifies pit dimensions as 0.5 m × 0.5 m × 0.5 m (50 cm × 50 cm × 50 cm) *[Source: Banana_disease_detection.pdf, Section 5.4, p. 4]*40.

- **Pit Solarization:** Dug pits must be left open to atmospheric solar radiation for 15 to 20 days (or 2 weeks) prior to refilling to desiccate soil-borne pathogens, kill harmful insect larvae, and promote soil aeration *[Source: Banana Expert System, Field Prep; Banana_disease_detection.pdf, Section 5.4, p. 4]*441.

- **Pit Refilling Mixture:** Pits are refilled with topsoil thoroughly mixed with 10 kg well-decomposed FYM (10–15 kg in wetlands), 250 g Neem cake, 20 g Carbofuran 3G (or Thimet/Furadon), and 250 g Superphosphate *[Source: Banana Expert System, Field Prep / TC Do's #6; Banana_disease_detection.pdf, Section 5.4, p. 4]*. In nematode-endemic sites, nematicides and fumigants are incorporated into pit soil *[Source: Banana Expert System, Field Prep]*4.

1.1.6 6. Soil Constraints and Reclamation Practices

Saline, Alkaline, and Sodic Soil Management (pH $>$ 7.5 or $>$ 8.0) In saline-alkali soils where pH exceeds 7.5 or 8.0, banana plants suffer from root suppression, marginal chlorosis/necrosis of older leaves, and yield reductions up to 30% *[Source: Banana Expert System, Contingency Problems; Banana_disease_detection.pdf, Section 5.4, p. 4]*4142.

- **Reclamation Measure:** Modify pit refilling mixture by incorporating organic matter (decomposed press mud or FYM) and Gypsum *[Source: Banana Expert System, Field Prep / TC Do's #3; Banana_disease_detection.pdf, Section 5.4, p. 4]*. Apply Gypsum @ 0.5 kg/plant + 15 kg FYM/plant along with 120% of recommended Potassium *[Source: Banana Expert System, Contingency Problems]*42.

- **Porosity and Drainage Improvement:** Addition of perlite improves soil porosity and aeration *[Source: Banana Expert System, Field Prep]*4. Deep drainage channels must be constructed to leach excess sodium ions and soluble salts out of the root zone *[Source: Banana Expert System, Contingency Problems]*42.

Acidic Soil Reclamation In acidic soils, apply soil amendments such as Dolomite ($MgCO_3$) or Limestone ($CaCO_3$) during land preparation *[Source: Banana.pdf, p. 14; Banana_disease_detection.pdf, Section 6, p. 5]*2325. Use Triple Superphosphate (TSP) or DAP instead of acidic rock phosphate sources *[Source: Banana_disease_detection.pdf, Section 6, p. 5]*25. For Panama wilt (*Fusarium oxysporum f. sp. cubense*) prone acidic soils, apply Quick Lime or Lime @ 1–2 kg/pit around the base with water soaking *[Source: Banana_disease_1.pdf, p. 2; Banana_disease_detection.pdf, Section 13.1, p. 31]*4344.

Poor Drainage, Waterlogging, and Flood Reclamation Waterlogging exacerbates Panama wilt and bacterial soft rot *[Source: Banana_disease_1.pdf, p. 1; Banana_disease_detection.pdf, Section 13.5, p. 35]*4546. In flood-affected or poorly drained fields,

open deep drainage trenches between rows *[Source: Banana Expert System, Contingency Problems]*47. To assist recovery, apply foliar sprays of 2% K_2SO_4, followed after 15 days by 1% KNO_3 *[Source: Banana Expert System, Contingency Problems]*47.

Biological Soil Disinfection and Solarization In fields contaminated with soilborne wilt or nematodes, flood/inundate the field with water for 1 to 3 months prior to planting *[Source: Banana_disease_detection.pdf, Section 13.1, p. 31]*44. Alternatively, spread 500–1,000 kg/acre of paddy or maize straw, flood for 20–30 days, or cover tilled/irrigated soil with plastic mulch for 6 to 8 weeks for soil solarization *[Source: Banana_disease_detection.pdf, Section 12.7 / 13.1, p. 28, 31]*4448. Sowing Sunnhemp seeds, growing for 45 days, and executing *in situ* green manure incorporation suppresses soil pathogens *[Source: Banana_disease_detection.pdf, Section 13.1, p. 31]*44.

1.1.7 7. Cultivar, Density, Planting System, and Regional Variations

Planting Systems and Density Considerations Plant population depends on cultivar growth habit, topography, and soil fertility *[Source: Banana Expert System, Spacing]*49.

- **Standard Spacing:** For cultivar 'Grand Nain', 6 ft × 5 ft (1.8 m × 1.5 m), 6 ft × 6 ft, or 7 ft × 5 ft is ideal; spacing below 5 ft × 5 ft is unadvisable *[Source: Banana Expert System, TC Do's #7]*5. General seasonal spacings in Maharashtra are 1.5 m × 1.5 m, 2.0 m × 2.0 m, or 2.5 m × 2.5 m (Kharif) and 1.5 m × 1.2 m or 1.5 m × 1.37 m (Rabi) *[Source: Banana_disease_detection.pdf, Section 5.2, p. 4]*40.
- **High-Density Planting (HDP):** Accommodates 4,444 to 5,555 plants/ha *[Source: Banana_disease_detection.pdf, Section 5.3, p. 4]*40. Planting 3 suckers/pit at 1.8 m × 3.6 m (4,600 plants/ha) for Cavendish bananas, or 2 suckers/pit at 2.0 m × 3.0 m (5,000 plants/ha) for 'Nendran' is recommended *[Source: Banana_disease_detection.pdf, Section 5.3, p. 4]*40.

Regional Planting Seasons

- **Maharashtra:** Kharif (June–July); Rabi (October–November) *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*50.
- **Tamil Nadu:** February–April and November–December *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*50.
- **Kerala:** Rainfed crop (April–May); Irrigated crop (August–September) *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*50.
- **South India (General):** Any time of year except severe summer; ideal time is post-monsoon (October–November) or February–March with assured irrigation *[Source: Banana Expert System, Spacing / Pit Method]*3949.
- **North India:** July–August *[Source: Banana Expert System, Pit Method]*39.

1.1.8 8. Operational Safety, Chemical Precautions, and Environmental Protection

Herbicide Safety and Application Restrictions For pre-emergence weed control, Diuron 80% WP @ 0.8 kg/acre (1 kg a.i./ha) or Glyphosate @ 2 kg a.i./ha is applied *[Source: Banana.pdf, p. 14; Banana_Intercultural_Operations_2.pdf, p. 1].* Pre-emergent or soil-incorporated herbicides must **never** be applied to dry soils *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38].* Herbicides must **never** be applied through irrigation water or mixed directly with soil, sand, or urea *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38].* Sprayers must be equipped with flat fan or floodjet nozzles operated at 15–20 psi *[Source: Banana.pdf, p. 41, 43].*

Soil-Applied Insecticide and Nematicide Handling When applying granular chemicals into pits (e.g., Carbofuran 3G @ 20–25 g/pit, Phorate 10G @ 10 g/pit, or Thimet/Furadon), operators must wear full protective clothing (coveralls, rubber boots, gloves, and mask) *[Source: Banana Expert System, TC Do's #6; Banana.pdf, p. 43–44; Banana_pests_-3.pdf, p. 2].* Equipment must be calibrated prior to use *[Source: Banana.pdf, p. 44]*57. Pesticide application must be terminated 7 to 15 days prior to fruit harvest to ensure pesticide residue-free export produce *[Source: Banana.pdf, p. 38, 42].*

1.2 Subtopic: Planting Material Selection, Variety Characteristics, Tissue Culture, and Pre-Planting Sucker Treatment

1.2.1 1. Commercial Banana Cultivars and Genetic Background

Modern cultivated bananas (*Musa spp.*) have evolved from two wild diploid species—*Musa acuminata* (A genome) and *Musa balbisiana* (B genome)—and their naturally occurring interspecific hybrids *[Source: Banana.pdf, p. 1].* Commercial production for domestic and export markets relies on specific cultivars adapted to local climate, soil conditions, and disease pressures *[Source: Banana.pdf, p. 1, 4; Banana_disease_detection.pdf, Section 2, p. 1].*

Key Commercial Cultivars and Genome Groups

- **Grand Nain (AAA - Cavendish Subgroup):** Highly preferred for export quality production due to uniform high yield, cylindrical bunch shape, and market demand *[Source: Banana Expert System, TC Do's #7; Banana_disease_detection.pdf, Section 5.3, p. 4].* Ideal spacing is 1.8 m × 1.5 m (6 ft × 5 ft) *[Source: Banana Expert System, TC Do's #7].*
- **Robusta and Dwarf Cavendish (AAA):** Widely grown in garden lands and biennial plantations; susceptible to Panama wilt (*Fusarium oxysporum f. sp. cubense*) *[Source: Banana Expert System, Pit Method; Banana.pdf, p. 14].*
- **Rasthali (AAB - Silk):** Highly prized dessert cultivar; extremely susceptible to Panama wilt and susceptible to viral diseases *[Source: Banana Expert System, Pre-treatment of suckers / Calendar 5th Month].*

- **Poovan (AAB - Mysore):** Commercial cultivar tolerant to Panama wilt; grown across diverse planting densities *[Source: Banana Expert System, Spacing / HDP; Banana.pdf, p. 14].*
- **Nendran (AAB - Plantain):** Major cooking and processing cultivar; sensitive to stem weevil (*Odoiporus longicollis*); highly suited for high-density paired planting (2 m × 3 m) *[Source: Banana Expert System, Calendar 5th Month; Banana_-disease_detection.pdf, Section 5.3, p. 4].*
- **Karpuravalli (ABB):** Dual-purpose cultivar exhibiting high tolerance to soil salinity, drought, nematodes, and Fusarium wilt *[Source: Banana Expert System, Pit Method; Banana_disease_detection.pdf, Section 12.7, p. 28].*
- **Monthan (ABB) and Virupakshi (AAB):** Cooking and hill banana cultivars susceptible to wilt disease, requiring strict corm pre-treatment *[Source: Banana Expert System, Pre-treatment of suckers / Calendar 5th Month].*
- **Other Regional Cultivars:** Pachanadan, Red Banana, Kadali, Kunnan, Poomkalli, Malbhog, Champa, Adukkar, and Poyo *[Source: Banana Expert System, Calendar 5th Month; Banana.pdf, p. 14; IPM-Banana-Revised-Sept2011.pdf, p. 14].*

Cultivar Disease Susceptibility and Selection Matrix

- **Panama Wilt Susceptible Cultivars (Avoid in endemic fields):** Robusta, Rasthali, Karpooravally, Malbhog, Champa, Adukkar, Monthan, Ney Poovan, and Pachanadan *[Source: Banana Expert System, Calendar 5th Month; Banana.pdf, p. 14].*
- **Panama Wilt Tolerant / Less Susceptible Cultivars:** Poovan, Kadali, Kunnan, and Poomkalli *[Source: Banana.pdf, p. 14].*
- **Nematode Tolerant Cultivars:** Karpuravalli, Monthan, Nattupoovan, and Kunnan *[Source: Banana_disease_detection.pdf, Section 12.7, p. 28].*

1.2.2 2. Types and Specifications of Planting Materials

Banana plantations are established using conventional vegetative suckers (rhizomes/corms) or micropropagated tissue culture plants *[Source: Banana Expert System, Planting Material; Banana_disease_detection.pdf, Section 5, p. 3].*

Sword Suckers vs. Water Suckers

- **Sword Suckers (Preferred):** Lateral shoots originating from healthy, disease-free mother corms with a strong, well-developed broad base tapering to a narrow tip, bearing sword-shaped leaves in early stages *[Source: Banana_disease_detection.pdf, Section 5, p. 3].* Sword suckers establish rapidly, exhibit high vegetative vigour, and shoot early bunches with uniform maturity *[Source: Banana_disease_-detection.pdf, Section 5, p. 3].*

- **Water Suckers (Unsuitable):** Broad-leaved early offshoots attached weakly to the upper part of the corm with small, poorly developed underground rhizomes; they produce weak plants with low yield potential and must be avoided *[Source: Banana Expert System, TC Do's #3; Banana_disease_detection.pdf, Section 5, p. 3].*

- **Sucker Size and Weight Requirements:** For conventional planting, select suckers weighing between 500g and 1.5kg (or corms > 500 g) *[Source: Banana_disease_detection.pdf, Section 13.5, p. 35].* For bacterial soft rot (*Erwinia carotovora*) endemic areas, use rhizomes with dead central buds and active lateral buds to prevent pathogen invasion through central vascular tissues *[Source: Banana_disease_detection.pdf, Section 13.5, p. 35].*

Tissue Culture (TC) Planting Material Standards Tissue culture plants produced via shoot tip micropropagation provide uniform, disease-free planting stock with high yield potential *[Source: Banana Expert System, Pre-treatment of suckers / TC Do's; Banana.pdf, p. 39].*

- **Planting Standard:** Certified TC plants must possess 5 to 6 fully expanded green leaves, a robust root system, and a sturdy stem base before field transplanting *[Source: Banana Expert System, Pre-treatment of suckers; Banana_disease_detection.pdf, Section 5, p. 3].*

- **Virus Indexing and Nursery Quarantine:** TC mother plants must undergo compulsory virus indexing using Enzyme-Linked Immunosorbent Assay (ELISA) or Dot-Immunobinding Assay (DIBA) with specific antiserums to detect Banana Bunchy Top Virus (BBTV), Banana Streak Virus (BSV), and Banana Bract Mosaic Virus (BBrMV) *[Source: IPM-Banana-Revised-Sept2011.pdf, p. 13–14].*

- **Viral Latency and Shoot Tip Culture Limitations:** Shoot tip culture does **not** eliminate Banana Streak Virus (BSV) or Banana Bunchy Top Virus (BBTV) if present in integrated genomic sequences; tissue culture stress can trigger episomal viral activation *[Source: Banana_disease_1.pdf, p. 3; IPM-Banana-Revised-Sept2011.pdf, p. 13–14].* Mother plants must be maintained at 22°C in vector-proof net houses for 2 to 3 months to monitor symptom expression prior to micropropagation *[Source: IPM-Banana-Revised-Sept2011.pdf, p. 14].*

1.2.3 3. Pre-Planting Sucker Preparation, Paring, and Sanitary Protocols

Conventional suckers extracted from mother fields harbor corm weevils (*Cosmopolites ordenarius / Odoiporus longicollis*), plant-parasitic nematodes (*Radopholus similis*, *Pratylenchus coffeae*, *Meloidogyne* spp.), and soil-borne fungal pathogens *[Source: Banana.pdf, p. 11, 14; Banana_disease_detection.pdf, Section 12.7, p. 25].*

Paring and Trimming (Mechanical Sanitation)

1. **Root and Corm Trimming:** Trim all fibrous roots and decayed outer tissues from the corm using a sharp knife *[Source: Banana Expert System, Pre-treatment of suckers].* Cut the pseudostem cleanly leaving 20 cm length attached to the upper corm *[Source: Banana Expert System, Pre-treatment of suckers].*

2. **Paring for Nematodes and Weevils:** Pare off the outer skin and dark discolored/lesioned cortical tissue until only clean, white, healthy corm tissue remains exposed *[Source: Banana_disease_detection.pdf, Section 12.7, p. 25; IPM-Banana-Revised-Sept2011.pdf, p. 10]*.

Thermotherapy (Hot Water Treatment) Submerge trimmed and pared suckers in a hot water bath maintained precisely at 50°C to 55°C for 20 minutes prior to chemical treatment *[Source: Banana_disease_detection.pdf, Section 12.5, 12.7, p. 25]*. This heat treatment eradicates endoparasitic burrowing nematodes (*Radopholus similis*) and thrips inhabiting the corm layers without impairing bud viability *[Source: Banana_disease_-detection.pdf, Section 12.5, 12.7, p. 25]*.

Solar Desiccation In large corms, store pared corms exposed to direct, bright sunlight for 14 days (2 weeks) prior to planting to desiccate hidden insect eggs and promote healing of cut surfaces *[Source: Banana.pdf, p. 14; Banana_disease_detection.pdf, Section 11, p. 38]*.

1.2.4 4. Pre-Planting Chemical Disinfection and Pralinage Protocols

Following paring and heat treatment, suckers must undergo chemical or bio-agent dipping to protect rhizomes during initial root emergence *[Source: Banana Expert System, Pre-treatment of suckers; Banana.pdf, p. 14]*.

Procedures, Dosages, and Reported Disagreements

- **Pralinage (Mud Slurry Method):** Dip trimmed corms in a thick mud slurry consisting of 4 parts clay mixed with 5 parts water. Sprinkle 40 g Carbofuran 3G granules per sucker over the wet slurry to coat the corm uniformly before planting *[Source: Banana Expert System, Pre-treatment of suckers]*.
- **Chemical Dipping for Nematodes and Weevils (Dipping vs. Pralinage Disagreements):**
  - *Banana Expert System* recommends dipping corms in 0.75% Monocrotophos solution, followed by shade drying for at least 24 hours *[Source: Banana Expert System, Pre-treatment of suckers]*.
  - *Banana_pests_3.pdf* specifies dipping suckers in 0.5% Monocrotophos solution for 30 minutes *[Source: Banana_pests_3.pdf, p. 1]*.
  - *Banana.pdf* specifies dipping suckers in Carbofuran 3% CG @ 33 g/sucker for rhizome weevil control, or 50 g/sucker for nematode control, followed by shade drying for 72 hours *[Source: Banana.pdf, p. 14]*.
  - *IPM-Banana-Revised-Sept2011.pdf* specifies dipping suckers for 30 minutes in a combined solution of Carbendazim 0.2% (2 g/L) + Monocrotophos 0.05% (0.5 ml/L) *[Source: IPM-Banana-Revised-Sept2011.pdf, p. 10]*.
- **Fungal Wilt Disinfection (Fusarium and Susceptible Cultivars):** For wilt-susceptible varieties ('Rasthali', 'Monthan', 'Virupakshi'), dip trimmed corms for 5 minutes in 0.1% Emisan solution (1 g/L water) or 0.2% Carbendazim solution

[Source: Banana Expert System, Pre-treatment of suckers; IPM-Banana-Revised-Sept2011.pdf, p. 10].

- **Bacterial Soft Rot Disinfection (*Erwinia*):** Dip or drench suckers in 2% Bleaching Powder solution or 10% Sodium Hypochlorite solution prior to field establishment *[Source: Banana_disease_detection.pdf, Section 13.5, p. 35–36].*

Biological Sucker Inoculation As an alternative or supplement to chemical pralinage, inoculate pared suckers with beneficial bio-control organisms at planting *[Source: Banana Expert System, Spacing; Banana.pdf, p. 14]*:

- Apply *Pseudomonas fluorescens* @ 25 g/plant or *Trichoderma viride* / *T. harzianum* @ 20 g/plant mixed with 500 g Neem cake per pit *[Source: Banana Expert System, Spacing; Banana_disease_detection.pdf, Section 12.7, p. 28].*
- For root-knot nematodes (*Meloidogyne* spp.), incorporate 2 kg FYM enriched with *Trichoderma harzianum* and *Purpureocillium lilacinus* (*Paecilomyces lilacinus*) into the planting pit *[Source: IPM-Banana-Revised-Sept2011.pdf, p. 12].*
- Apply vesicular-arbuscular mycorrhizal fungus *Glomus mosseae* with 500 g Neem cake per plant to enhance phosphorus uptake and reduce root necrosis *[Source: IPM-Banana-Revised-Sept2011.pdf, p. 12].*

1.2.5 5. Operational Safety, Chemical Precautions, and Environmental Handling

Handling Granular Nematicides and Systemic Insecticides Chemicals used for sucker treatment (Carbofuran 3G/3% CG, Monocrotophos, Phorate 10G) are highly toxic synthetic organophosphates and carbamates *[Source: Banana Expert System, TC Do's #6; Banana.pdf, p. 43–44].*

- **Personal Protective Equipment (PPE):** Operators mixing slurry, dipping suckers, or applying granules into pits must wear full protective gear, including rubber gloves, face masks, protective goggles, long-sleeved coveralls, and rubber boots *[Source: Banana.pdf, p. 43–44].*
- **Shade Drying Safety:** Suckers treated with Monocrotophos or Carbofuran slurry must be shade dried in locked, ventilated structures away from livestock, children, and water sources for 24 to 72 hours *[Source: Banana Expert System, Pre-treatment of suckers; Banana.pdf, p. 14].*
- **Disposal and Environmental Protection:** Wastewater from sucker dipping tanks must never be drained into farm canals, ponds, or open streams to prevent aquatic toxicity and groundwater contamination *[Source: Banana.pdf, p. 38, 44].* Empty chemical bags and containers must be triple-rinsed and disposed of according to national hazardous waste regulations *[Source: Banana.pdf, p. 44].*

1.3 Subtopic: Planting Systems, Spacing, Density, Layout Methods, Irrigation Management, and Fertigation Schedules

1.3.1 1. Seasonal Planting Timings and Climate Constraints

Banana planting timing is governed by regional temperature regimes, rainfall distribution, and irrigation availability *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4; Banana Expert System, Spacing and Planting]*. Planting is strictly recommended during post-monsoon or early summer months and should be avoided during severe cold weather (temperatures below 15°C), extreme summer heat (above 40°C), or during heavy, continuous rains when soils remain saturated *[Source: Banana Expert System, TC Do's #6; IPM-Banana-Revised-Sept2011.pdf, p. 86]*.

State-Wise Commercial Planting Calendar

- **Maharashtra:** Kharif season (June–July) and Rabi season (October–November) *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*.
- **Tamil Nadu:** Early planting (February–April) and late planting (November–December) *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*.
- **Kerala:** Rainfed production (April–May) and irrigated crop establishment (August–September) *[Source: Banana_disease_detection.pdf, Section 5.1, p. 4]*.
- **South India (General):** Year-round planting except peak summer; ideal post-monsoon period is October–November, or February–March under assured irrigation *[Source: Banana Expert System, Spacing / Pit Method]*.
- **North India:** July–August coinciding with the onset of the South-West monsoon *[Source: Banana Expert System, Pit Method]*.

1.3.2 2. Planting Systems, Geometrical Layouts, and Field Methods

Selecting the appropriate geometrical planting layout depends on canopy light interception, mechanization requirements, field topography, and irrigation delivery infrastructure *[Source: Banana Expert System, System of Planting]*.

Geometrical Layout Systems

- **Square System:** The most widely adopted layout, where plants occupy the corners of a square (typically spaced at 1.8 m × 1.8 m) *[Source: Banana Expert System, Square System]*. It permits intercultural operations and tractor tillage in two directions and allows short-duration intercrops in central spaces *[Source: Banana Expert System, Square System]*.
- **Triangular System:** Recommended specifically for tissue culture plants *[Source: Banana Expert System, Triangular System]*. Inter-row spacing is maintained at 1.5m with intra-row plant spacing at 1.8m, positioning plants in even-numbered rows midway relative to odd-numbered rows *[Source: Banana Expert System, Triangular System]*. Although each plant receives equal light interception, it accommodates slightly fewer plants per hectare than the square system *[Source: Banana Expert System, Triangular System]*.

- **Single Row System:** Characterized by close intra-row spacing and wide inter-row corridors *[Source: Banana Expert System, Single Row System]*. This architecture enhances light penetration and air circulation through the canopy, accelerating the drying of wet leaves and suppressing fungal foliage diseases (such as Sigatoka leaf spot), though total plant population per unit area is reduced *[Source: Banana Expert System, Single Row System]*.

- **Paired Row System:** Inter-line distance between twin rows is set at 0.90m to 1.20m, while plant-to-plant distance along the line is 1.2m to 2.0m *[Source: Banana Expert System, Paired Row System]*. This layout creates wide working corridors, facilitating tractor-drawn intercultural implements and substantially lowering drip lateral piping costs *[Source: Banana Expert System, Paired Row System]*.

Field Planting Methods and Source Disagreements

- **Pit Method:** Standard practice in garden land systems and biennial plantations (for cultivars such as 'Dwarf Cavendish', 'Robusta', 'Rasthali', 'Poovan', and 'Karpuravalli') *[Source: Banana Expert System, Pit Method]*. Pits are excavated, solarized, and refilled with topsoil, FYM, and soil amendments *[Source: Banana Expert System, Field Prep; Banana_disease_detection.pdf, Section 5.4, p. 4]*. Suckers or tissue culture plants are placed upright in the pit center, ensuring the soil is compacted firmly around the base to eliminate hollow air pockets *[Source: Banana Expert System, Planting]*. Tissue culture plants are planted at ground level, while conventional suckers are set with pseudostems 2–5cm below ground level *[Source: Banana Expert System, Planting; Banana_disease_detection.pdf, Section 5.4, p. 4]*. Deep planting ($>$ 15–20 cm) must be avoided to prevent stem rot and delayed shoot emergence *[Source: Banana_disease_detection.pdf, Section 5.4, p. 4; Banana.pdf, p. 39]*.

  - *Source Disagreement on Pit Dimensions: Banana Expert System (Field Prep)* specifies pit dimensions of 45 cm $\times$ 45 cm $\times$ 45 cm *[Source: Banana Expert System, Field Prep]*; *Banana Expert System (Pit Method)* specifies 60 cm $\times$ 60 cm $\times$ 60 cm *[Source: Banana Expert System, Pit Method]*; and *Banana_disease_detection.pdf* reports pit dimensions of 0.5 m $\times$ 0.5 m $\times$ 0.5 m *[Source: Banana_disease_detection.pdf, Section 5.4, p. 4]*.

- **Furrow Method:** Adopted in annual planting systems across Maharashtra and Gujarat *[Source: Banana Expert System, Furrow Method; Banana_disease_detection.pdf, Section 5.4, p. 4]*. Ridgers or manual ploughs open 30–40cm deep furrows across the field *[Source: Banana Expert System, Furrow Method]*. Suckers are set at recommended intervals along furrows, packed with FYM and soil. During the rainy season, earthing up is executed to raise ridges around pseudostems to prevent waterlogging; during dry summer and winter months, plants remain in furrows to maximize moisture retention *[Source: Banana Expert System, Furrow Method; Banana_Intercultural_Operations_2.pdf, p. 2]*.

- **Trench Method:** Practiced in wetland cultivation across the Cauvery Delta region of Tamil Nadu *[Source: Banana Expert System, Trench Planting; Banana_disease_detection.pdf, Section 5.4, p. 4]*. Fields are puddled with gauge wheels and

drained for 24 hours *[Source: Banana Expert System, Trench Planting]*. Suckers are pressed directly into soft mud. At 7 days post-planting, 15cm deep trenches are cut between blocks of 4 to 6 plants *[Source: Banana Expert System, Trench Planting]*. These trenches are deepened monthly by 20–25cm, reaching a final depth of 60 cm by the third month to act as drainage canals during monsoons and organic cycling channels post-monsoon *[Source: Banana Expert System, Trench Planting]*.

1.3.3 3. Plant Density, High-Density Planting (HDP), and Spacing Standards

Standard Spacing Recommendations Plant density depends on cultivar stature, soil fertility, climate, and management intensity *[Source: Banana Expert System, Spacing; Banana_disease_detection.pdf, Section 5.2, p. 4]*.

- **Grand Nain (AAA):** Ideal standard spacing is 1.8 m × 1.5 m (6 ft × 5 ft), yielding 3,703 plants/ha *[Source: Banana Expert System, TC Do's #7]*. Alternative spacings include 1.8 m × 1.8 m (6 ft × 6 ft) or 2.1 m × 1.5 m (7 ft × 5 ft) *[Source: Banana Expert System, TC Do's #7]*. Spacings closer than 1.5 m×1.5 m (5 ft×5 ft) are unadvisable for Grand Nain in single-sucker systems *[Source: Banana Expert System, TC Do's #7]*.
- **Seasonal Spacing (Maharashtra):** Kharif crop spacing is set at 1.5 m × 1.5 m, 2.0 m × 2.0 m, or 2.5 m × 2.5 m; Rabi crop spacing is set closer at 1.5 m × 1.2 m or 1.5 m × 1.37 m *[Source: Banana_disease_detection.pdf, Section 5.2, p. 4]*.

High-Density Planting (HDP) Systems HDP accommodates 4,444 to 5,555 plants/ha to maximize radiation conversion efficiency and land use productivity *[Source: Banana_disease_detection.pdf, Section 5.3, p. 4; Banana Expert System, High Density Planting]*. Energy conversion efficiency is highest at 1.2 m×1.2 m spacing and lowest at 2.1 m×2.1 m spacing *[Source: Banana Expert System, High Density Planting]*.

- **Cavendish Subgroup (e.g., Robusta, Grand Nain):** Plant 3 suckers per pit at a wide spacing of 1.8 m×3.6 m, establishing 4,600 plants/ha *[Source: Banana_disease_detection.pdf, Section 5.3, p. 4; Banana Expert System, High Density Planting]*.
- **Nendran (AAB):** Plant 2 suckers per pit at 2.0 m × 3.0 m spacing, establishing 5,000 plants/ha *[Source: Banana_disease_detection.pdf, Section 5.3, p. 4]*.
- **HDP Agronomic Management:** HDP requires integration with short-duration cowpea intercropping, precise drip irrigation, and adjusted fertilizer regimes (applying 50%, 75%, or 100% of recommended NPK per plant base) to prevent reduction in finger length and weight *[Source: Banana Expert System, High Density Planting]*.

1.3.4 4. Irrigation Management and Water Regimes

Total Crop Water Requirement and Conventional Irrigation The total water requirement of a banana plant over its complete life cycle is 900mm to 1,200mm *[Source: Banana_disease_detection.pdf, Section 7, p. 7]*. Supplementary irrigation is essential during dry periods to maintain continuous soil moisture in the upper root zone *[Source:*

Banana_disease_detection.pdf, Section 7, p. 7]. Under conventional surface basin irrigation, water must be applied every 3 to 4 days during hot summer months and at 7 to 8 day intervals during cool winter months *[Source: Banana_disease_detection.pdf, Section 7, p. 8].* Flood irrigation and top-surface chemical washing must be avoided *[Source: Banana Expert System, TC Do's #11].*

Drip Irrigation System Standards Drip irrigation saves 30% to 40% water compared to surface flooding, improves water and nutrient use efficiency, accelerates vegetative growth, advances bunch emergence, and suppresses weed growth between rows *[Source: Banana_disease_detection.pdf, Section 7, p. 8].*

Table 2: Stage-Specific Daily Drip Water Requirement for Banana

S. No.	Crop Growth Stage	Duration (Weeks)	Water Qu
1	After Planting / Ratoon Establishment	1–4	4–6 *[Source: Banana_*
2	Juvenile Vegetative Phase	5–9	8–10 *[Source: Banana*
3	Critical Growth Stage	10–19	12 *[Source: Banana_*
4	Flower Bud Differentiation Stage	20–32	16–20 *[Source: Banana*
5	Shooting / Bunch Emergence Stage	33–37	20 and above* *[Source: Ba*
6	Bunch Development to Maturity Stage	38–50	20 and above* *[Source: Ba*
*Water quantity adjusted based on seasonal evapotranspiration rates.			

1.3.5 5. Fertigation Schedules and Nutrient Delivery

Applying water-soluble fertilizers through drip irrigation (fertigation) prevents nutrient leaching, gaseous volatilization of nitrogen, and phosphorus/potassium fixation in soil matrixes *[Source: Banana.pdf, p. 15].* Drip fertigation saves 25% of the Recommended Dose of Fertilizers (RDF) *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*

Daily Fertigation Program for Tissue Culture Banana (1,200 Plants/Acre Density) Basal soil application of Superphosphate @ 250 g/plant is incorporated into the soil during pit refilling prior to initiating fertigation *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*

Table 3: Daily Fertigation and Water Schedule for Tissue Culture Banana (1,200 Plants/Acre)

Days After Planting	Urea (kg/day/acre)	Superphosphate	MOP / KCl (kg/day/acr
30–75 (45 Days)	0.60	250 g/plant (Basal)	0.90
75–120 (45 Days)	1.20	—	1.50
120–165 (45 Days)	2.40	—	3.60
165–210 (45 Days)	3.00	—	4.80
210–255 (45 Days)	2.70	—	3.60
255–270 (15 Days)	2.40	—	3.00

Weekly Fertigation Program (Gram/Plant/Week Standard)

- **Weeks 9 to 18 (10 Weeks):** Apply 15g Urea + 8.0g Muriate of Potash (MOP) per plant per week *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*
- **Weeks 19 to 30 (12 Weeks):** Apply 10g Urea + 10g MOP per plant per week *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*
- **Weeks 31 to 40 (10 Weeks):** Apply 7.0g Urea + 12g MOP per plant per week *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*
- **Weeks 41 to 46 (5 Weeks):** Apply 0g Urea + 10g MOP per plant per week *[Source: Banana_disease_detection.pdf, Section 6, p. 5].*

1.3.6 6. Operational Precautions, Drainage Management, and Winter Protection

Pre-Harvest Irrigation Termination Irrigation must be stopped completely 7 days (1 week) prior to bunch harvesting *[Source: Banana Expert System, TC Do's #13].* This practice firms up field roads, facilitates efficient labor movement, prevents fruit splitting during cutting, and reduces latex staining during harvest transportation *[Source: Banana Expert System, TC Do's #13].*

Drainage Channels and Waterlogging Management Providing adequate deep drainage channels to drain stagnant water out of the root zone is mandatory *[Source: Banana_disease_detection.pdf, Section 7, p. 8; IPM-Banana-Revised-Sept2011.pdf, p. 87].* Waterlogging increases soil humidity, creates anaerobic conditions, causes corm rotting, and accelerates the spread of soil-borne pathogens (*Erwinia carotovora* and *Fusarium oxysporum*) *[Source: Banana Expert System, Contingency Problems; Banana_disease_detection.pdf, Section 13.5, p. 35].* In flooded plantations, open deep trenches between rows and apply foliar sprays of 2% K_2SO_4 followed by 1% KNO_3 after 15 days to assist root recovery *[Source: Banana Expert System, Contingency Problems].*

Chilling Injury Mitigation In northern and western growing regions where night temperatures drop below 12°C or 5°C during winter, plantations suffer from chilling injury, leaf chlorosis, and delayed bunch emergence *[Source: Banana Expert System, Contingency Problems; Banana_disease_detection.pdf, Section 2, p. 1].*

- **Irrigation Regime:** Maintain adequate soil moisture during winter spells; never allow fields to dry out during cold waves *[Source: Banana Expert System, TC Do's #11].*
- **Bunch Protection:** Cover developing bunches with 100-gauge perforated polythene sleeves featuring 6% vent openings to maintain internal microclimate temperatures and prevent skin discoloration *[Source: Banana Expert System, Contingency Problems].*

1.4 Subtopic: Intercultural Operations, Canopy Management, Bunch Care, and Harvest Management

1.4.1 1. Integrated Weed Management (IWM)

Weed competition during the first 4 months post-planting severely restricts banana root architecture and vegetative growth *[Source: Banana_Intercultural_Operations_2.pdf, p. 1; Banana_disease_detection.pdf, Section 11, p. 38]*. A weed-free basin must be maintained routinely around each pseudostem *[Source: Banana Expert System, Weeding / TC Do's #15; Banana.pdf, p. 14]*.

Cultural and Mechanical Weeding Deep spading 4 times a year is effective in controlling weed growth in garden lands *[Source: Banana_Intercultural_Operations_-2.pdf, p. 1]*. Regular mowing of inter-row vegetation or shallow inter-row ploughing is recommended during vegetative and reproductive stages to deplete the soil weed seed bank *[Source: Banana.pdf, p. 14, 18]*. Cover cropping with short-duration leguminous crops (such as cowpea or sunnhemp) suppressed weeds while fixing atmospheric nitrogen *[Source: Banana Expert System, Field Prep; Banana_Intercultural_Operations_2.pdf, p. 1]*.

Herbicide Regimes and Safety Restrictions For pre-emergence control of annual grasses and broad-leaved weeds without impairing crop yield or fruit quality:

- Apply Diuron @ 1kg a.i./ha (or 0.8kg/acre of 80% WP) or Glyphosate @ 2 kg a.i./ha *[Source: Banana.pdf, p. 14; Banana_Intercultural_Operations_2.pdf, p. 1]*.
- **Application Restrictions:** Pre-emergence or soil-incorporated herbicides must **never** be applied to dry soils *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38]*. Herbicides must **never** be injected into irrigation streams or mixed directly with sand, soil, or urea fertilizers *[Source: Banana.pdf, p. 38; Banana_disease_detection.pdf, Section 11, p. 38]*. Operators must use sprayers equipped with flat fan or floodjet nozzles operated at low pressures (15–20 psi) to prevent spray drift onto lower green leaves *[Source: Banana.pdf, p. 41, 43]*.

1.4.2 2. Desuckering and Ratooning Management

Uncontrolled sucker development diverts water, mineral nutrients, and photosynthates away from the main bearing pseudostem, reducing bunch weight and delaying shooting *[Source: Banana Expert System, Desuckering; Banana_Intercultural_Operations_2.pdf, p. 1–2]*.

Desuckering Frequency and Methods Removal of surplus suckers (desuckering) is one of the most critical intercultural operations *[Source: Banana_disease_detection.pdf, Section 8.4, p. 10]*.

- **Frequency:** Conduct desuckering at monthly intervals in garden lands and wetlands *[Source: Banana Expert System, Desuckering]*. Removal of unwanted suckers with a portion of the corm at 5–6 week intervals accelerates shooting and maximizes

bunch yield *[Source: Banana_disease_detection.pdf, Section 8.4, p. 10; Banana_-Intercultural_Operations_2.pdf, p. 2].*

- **Execution Methods:** Cut the sucker flush at ground level, gouge out the central heart/apical growing point without detaching the sucker, or pour 2–3 ml of kerosene oil directly into the growing point of the excised sucker core *[Source: Banana Expert System, Desuckering; Banana_disease_detection.pdf, Section 8.4, p. 10].*

Ratooning and Sucker Selection For establishing ratoon crops, retain only one healthy, vigorous sword sucker per stool/mat *[Source: Banana Expert System, Desuckering].*

- **Ratoon Setting Protocol:** The selected sucker is headed back once at 2 months post-emergence to arrest apical height growth, encouraging rhizome enlargement *[Source: Banana Expert System, Desuckering].* At harvest of the plant crop, the selected ratoon sucker reaches optimum vigor for the next cycle *[Source: Banana Expert System, Desuckering].*
- **Hill Banana System:** Retain 2 bearing pseudostems and 2 follower suckers per clump, removing all extra offshoots *[Source: Banana Expert System, Desuckering].*

1.4.3 3. Canopy Management and Leaf Pruning

Foliage Sanitation and Pruning Benefits Pruning surplus, old, yellowed, or diseased leaves alters canopy microclimate temperature and light penetration, improves ventilation, suppresses Sigatoka leaf spot (*Mycosphaerella musicola / M. fijiensis*), and prevents leaf tearing against developing fingers *[Source: Banana Expert System, Pruning of Leaves; Banana_Intercultural_Operations_2.pdf, p. 2].* Dry leaves should be cut flush with the pseudostem, removed from the field, and burned or composted *[Source: Banana Expert System, TC Do's #15; IPM-Banana-Revised-Sept2011.pdf, p. 45].*

Critical Leaf Threshold at Flowering To ensure complete bunch maturation, a minimum of 6 to 8 healthy, fully expanded green leaves must be preserved on the plant at shooting/flowering *[Source: Banana Expert System, Removal of withered styles; Banana_disease_detection.pdf, Section 8.5, p. 10].* Excessive leaf removal or severe foliage damage prior to bunch initiation substantially decreases bunch weight and shortens the post-harvest green life of harvested fruit *[Source: Banana Expert System, Removal of withered styles].*

Mattacking (Post-Harvest Pseudostem Management) and Source Disagreements After bunch harvest, the pseudostem is cut back *[Source: Banana Expert System, Mattacking; Banana_disease_detection.pdf, Section 8.6, p. 10].*

- **Source Disagreement on Post-Harvest Pseudostem Retention:** *Banana Expert System (Mattacking / TC Do's #18)* mandates cutting the pseudostem at a height of 0.6–1.0 m above ground level, leaving the stump intact so stored food reserves and nutrients leach down into the connected ratoon sucker *[Source: Banana Expert System, Mattacking / TC Do's #18].* Conversely, *Banana_disease_detection.pdf (Section 8.6, p. 10–11)* states that in zones endemic to banana stem weevil

(*Odoiporus longicollis*) and corm weevil (*Cosmopolites sordidus*), retaining standing stumps is strictly discouraged; instead, the pseudostem must be chopped flush at ground level and shredded or mulched into the soil to eliminate borer breeding sites *[Source: Banana_disease_detection.pdf, Section 8.6, p. 10–11].*

1.4.4 4. Earthing Up, Mulching, and Ground Maintenance

Earthing Up In furrow-planted fields, earthing up must be executed prior to the rainy season to raise soil ridges around pseudostems *[Source: Banana Expert System, Earthing up; Banana_Intercultural_Operations_2.pdf, p. 2].* This operation covers exposed upper corm zones, encourages deep secondary root anchoring, improves field drainage, and prevents crop lodging during heavy monsoons *[Source: Banana Expert System, Earthing up].* During dry summer and winter months, plants remain in furrows to optimize irrigation distribution *[Source: Banana Expert System, Furrow Method; Banana_Intercultural_-Operations_2.pdf, p. 2].*

Mulching Practices Applying organic mulch (such as dry sugarcane trash, paddy straw, or banana foliage) over basin surfaces conserves soil moisture, regulates soil temperature, increases feeder root density in upper soil layers, boosts water and nutrient use efficiency, suppresses weed growth, and increases banana fruit yields by 30% to 40% *[Source: Banana Expert System, Mulching; Banana_Intercultural_Operations_2.pdf, p. 1].*

1.4.5 5. Propping and Pseudostem Support

Propping is mandatory at bunch emergence in wind-prone areas (wind speeds > 30 km/hr) and for tall cultivars bearing heavy bunches *[Source: Banana Expert System, Propping; Banana_disease_detection.pdf, Section 8.7, p. 11].*

- **Support Structures:** Pseudostems are propped using bamboo, casuarina, or eucalyptus poles featuring an effective field life of 3–4 years *[Source: Banana Expert System, Propping; Banana_disease_detection.pdf, Section 8.7, p. 11].* Poles are tied forming a triangle or fork brace at the neck of the pseudostem below the bunch crown *[Source: Banana Expert System, Propping].* High-tensile polythene wires or coir ropes tied to border anchor posts can also be utilized *[Source: Banana Expert System, Propping].*

1.4.6 6. Bunch Care and Quality Enhancement Operations

Deflowering Deflowering consists of removing withered floral styles and perianth parts from the tips of young fingers immediately after hand opening *[Source: Banana Expert System, Removal of withered styles; Banana_disease_detection.pdf, Section 8.2, p. 10].* This operation removes dead organic tissue that serves as an entry vector for fungal pathogens such as cigar end rot (*Verticillium theobromae*) and tip rot (*Botryodiplodia theobromae*) *[Source: Banana Expert System, Removal of withered styles; Banana.pdf, p. 12].*

Denavelling (Male Bud Removal) Removal of the male flower bud (denavelling) is executed immediately after female phase completion (following the emergence of the last fruit hand and 1–2 neutral hands) *[Source: Banana Expert System, Denavelling; Banana_disease_detection.pdf, Section 8.3, p. 10].* The inflorescence rachis is cut 15–20 cm below the final hand *[Source: Banana_Intercultural_Operations_2.pdf, p. 2–3].* Denavelling checks the movement of photosynthates into an unwanted sink, promotes finger filling, advances fruit maturity, and increases total bunch weight *[Source: Banana Expert System, Denavelling; Banana_disease_detection.pdf, Section 8.3, p. 10].*

Bunch Thinning Remove 1 to 2 small, incomplete bottom hands from the tip of the developing bunch, retaining 7 to 8 uniform hands per bunch *[Source: Banana Expert System, Bunch Thinning].* This encourages uniform finger elongation, prevents spatial crowding, and aligns bunch grade with export packing specifications *[Source: Banana Expert System, Bunch Thinning].*

Peduncle Wrapping During peak summer months, the exposed main peduncle stalk connecting the bunch to the pseudostem is vulnerable to sunburn and tissue scorching *[Source: Banana Expert System, Peduncle Wrapping].* Scorched peduncle tissue provides an entry point for fungal and bacterial decay, causing peduncle rot, premature fruit drop, or uneven ripening *[Source: Banana Expert System, Peduncle Wrapping].* Wrapping the upper peduncle with the flag leaf or dry banana leaves insulates vascular tissues against solar radiation *[Source: Banana Expert System, Peduncle Wrapping; Banana_disease_detection.pdf, Section 8.1, p. 10].*

Bunch Covering (Sleeving) Covering developing bunches is a standard cultural requirement for export quality production *[Source: Banana Expert System, Bunch Covering; Banana_Intercultural_Operations_2.pdf, p. 3].*

- **Covering Materials:** Bunches are covered using dry banana leaves, gunny bags, or 100-gauge blue/white perforated polythene or polypropylene sleeves featuring 6% vent openings *[Source: Banana Expert System, Bunch Covering / Contingency Problems; Banana_disease_detection.pdf, Section 8.1, p. 10].*
- **Quality Benefits:** Sleeving increases fruit yield by 15% to 20%, protects fingers against cold chilling injury, sun scorching, dust accumulation, and fruit scarring caused by thrips (*Cheatanophothrips signipennis*) and scarring beetles (*Nodostoma subcostatum*) *[Source: Banana Expert System, Bunch Covering; Banana_Intercultural_Operations_2.pdf, p. 3; Banana_disease_detection.pdf, Section 8.1, p. 10].*
- **Inspection Protocol:** Regular visual checking under sleeves is essential to ensure moisture accumulation or secondary pest outbreaks are caught early *[Source: Banana_disease_detection.pdf, Section 12.5, p. 25].*

Tagging Bunches must be tagged with color-coded ribbons corresponding to the specific week of bunch shooting and sleeving *[Source: Banana Expert System, Tagging].* Color tagging enables precise harvesting based on visual maturity calculations *[Source: Banana Expert System, Tagging].*

1.4.7 7. Harvesting Protocols, Maturity Standards, and Post-Harvest Handling

Harvest Maturity Indicators Bunches are harvested when fruits reach three-quarters full or full maturity depending on target market distance *[Source: Banana Expert System, Harvesting; Banana_disease_detection.pdf, Section 10, p. 20]*.

- **Visual Indicators:** Prominent fruit angles disappear, becoming plump and rounded in cross-section; upper leaves turn yellow; and floral remnants drop off easily *[Source: Banana Expert System, Harvesting; Banana_disease_detection.pdf, Section 10, p. 20]*.
- **Crop Duration:** Harvesting typically occurs 90 to 120 days post-shooting depending on temperature and cultivar *[Source: Banana Expert System, Harvesting]*.

Pre-Harvest Operational Restrictions

- **Irrigation Suspension:** Stop irrigation completely 7 days (1 week) prior to bunch harvest to firm field paths, facilitate transport, prevent fruit splitting, and minimize latex staining during cutting *[Source: Banana Expert System, TC Do's #13]*.
- **Pesticide Pre-Harvest Interval (PHI):** Terminate all chemical pesticide applications 7 to 15 days prior to harvest to guarantee pesticide residue-free export produce *[Source: Banana.pdf, p. 38, 42]*.

Bunch Cutting and Transport Handling

- **Cutting Method:** Cut the main peduncle stalk cleanly 30 to 45cm above the first hand to provide a handling handle *[Source: Banana Expert System, Harvesting]*. Bunches must be lowered gently onto padded shoulders or foam cushions and loaded onto padded field trailers to prevent mechanical bruising *[Source: Banana.pdf, p. 20]*.
- **Field Shade and Temperature Limits:** Harvested bunches must be transferred immediately to shaded packing structures *[Source: Banana.pdf, p. 20]*. Direct exposure to solar heat causes rapid pulp degradation and severe shelf-life reduction *[Source: Banana.pdf, p. 20]*.
- **Cold Storage Threshold:** Fruit must be washed, drained, de-handed, graded, and packed into 13kg corrugated fiberboard cartons *[Source: Banana.pdf, p. 20]*. Mature-green export bananas can be stored for up to 3 weeks in ethylene-free air or up to 6 weeks in controlled atmospheres at a temperature of 14°C *[Source: Banana.pdf, p. 21]*. **Critical Storage Limit:** Storage or transport temperatures must **never** drop below 14°C, as lower temperatures induce irreversible chilling injury, skin darkening, and latex coagulatory failure *[Source: Banana.pdf, p. 20–21; Banana_disease_detection.pdf, Section 2, p. 1]*.